

Decision and Risk Analysis

Lecture 1: **Bayesian Thinking for Risk and Decision Analysis**

DAEN 427/ISEN 427/ISEN 627

Alexander Abuabara

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A decision problem begins with a **choice to be made** under certain **objectives** and **constraints**.

Risk and Uncertainty

- **Choice under risk:** Outcome probabilities are known (or can be estimated reliably).
- **Choice under uncertainty:** Outcome probabilities are unknown (or difficult to estimate).

Decision and Risk Analysis

1. **Modeling Decisions:** Define the available choices or actions.
2. **Modeling Risk:** Analyze situations where outcome probabilities are known.
3. **Modeling Uncertainty:** Analyze situations where outcome probabilities are unknown (or difficult to estimate).
4. **Modeling Preferences:** Represent how decision makers value outcomes and trade-offs.

Decision and Risk Analysis, Examples

1. **Modeling Decisions**

A student chooses between studying for an exam tonight or attending a social event.

2. **Modeling Risk**

A company launches a new product knowing there is a 70% chance of high demand and a 30% chance of low demand.

3. **Modeling Uncertainty**

A farmer decides what crop to plant without knowing how much rain will fall this season.

4. **Modeling Preferences**

A traveler chooses between a cheaper flight with a long layover and a more expensive direct flight.

Quiz 1

When flipping a fair coin, we say that “the probability of flipping Heads is 0.5.” How do you interpret this probability?

- a. If I flip this coin over and over, roughly 50% will be Heads.
- b. Heads and Tails are equally plausible.
- c. Both (a) and (b) make sense.

Quiz 2

An election is coming up and a pollster claims that candidate A has a 0.9 probability of winning. How do you interpret this probability?

- a. If we observe the election over and over, candidate A will win roughly 90% of the time.
- b. Candidate A is much more likely to win than to lose.
- c. The pollster's calculation is wrong. Candidate A will either win or lose, thus their probability of winning can only be 0 or 1.

Quiz 3

Consider two claims:

- 1 Zuofu claims that he can predict the outcome of a coin flip. To test his claim, you flip a fair coin 10 times and he correctly predicts all 10.
- 2 Kavya claims that she can distinguish natural and artificial sweeteners. To test her claim, you give her 10 sweetener samples and she correctly identifies each.

In light of these experiments, what do you conclude?

- a. You're more confident in Kavya's claim than Zuofu's claim.
- b. The evidence supporting Zuofu's claim is just as strong as the evidence supporting Kavya's claim.

Quiz 4

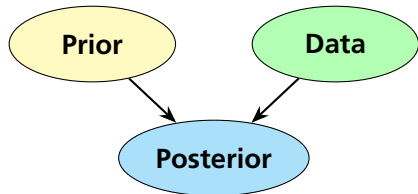
Suppose that during a recent doctor's visit, you tested positive for a rare disease. If you only get to ask the doctor one question, which would it be?

- a. What's the chance that I actually have the disease?
- b. If in fact I don't have the disease, what's the chance that I would've gotten this positive test result?

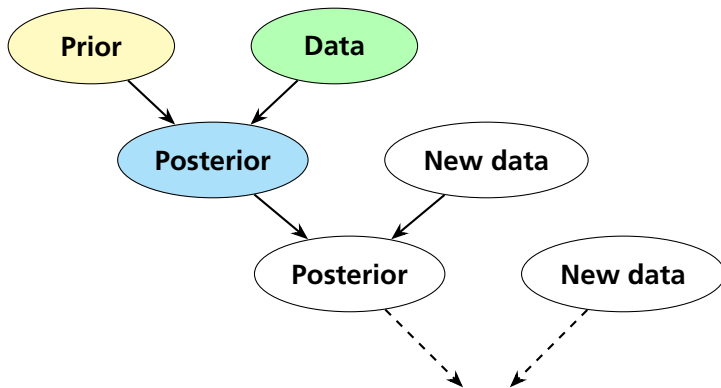
Bayesian Modeling of Decisions, Uncertainty, Risk, and Preferences

- Bayesian modeling provides a way to update beliefs as new information becomes available.
- It combines:
 1. **Prior beliefs**: what we know / assume before seeing new data.
 2. **Evidence/data**: new information relevant to the decision.
 3. **Posterior beliefs**: updated probabilities after incorporating the evidence.
- Bayesian modeling helps decision makers revise risk estimates and make better choices under uncertainty.
- Supporting learning from evidence and improving decisions over time.

A Bayesian Knowledge-Building Diagram



A Bayesian Knowledge-Building Diagram



Interpreting Probability

- The frequentist philosophy is so named for its interpretation of probability as the long-run relative frequency of a repeatable event.
- In the Bayesian philosophy, a probability measures the relative plausibility of an event.

Is this coin fair?

- **Frequentist:** "If the coin were fair, how surprising would these flips be?"
- **Bayesian:** "Given these flips, how much should I believe the coin is fair?"

One-Time Events

- Many important events are unrepeatable:
Big Bang, elections, weather and climate change, ...
- **Frequentist** probability can be awkward for one-time events because there is no long-run repetition.
- **Bayesian** probability is more flexible:
 - It expresses degree of belief.
 - It incorporates model, evidence, and uncertainty.
- A 0.9 probability of winning means the candidate is considered 9 times more likely to win than lose.

The Bayesian Balancing Act

- Bayesian and frequentist approaches interpret the same evidence differently:
 - **Frequentist:** Treats "10 out of 10" identically in all cases.
 - **Bayesian:** Plausibility on observed evidence given prior beliefs.

Both achieve 10 out of 10 correct in testing:

- Zuofu claims he can predict coin flips.
... is doubtful, predicting coin flips is highly implausible.
- Kavya claims she can distinguish natural / artificial sweeteners.
... is more believable.

Ask Questions!

- A **frequentist** analysis assesses the uncertainty of the observed data in light of an assumed hypothesis.
- A **Bayesian** analysis assesses the uncertainty of the hypothesis in light of the observed data.

Scenario: Population 100 people, 4 have the disease, 96 do not.

	Test +	Test –	Total
Disease	3	1	4
No disease	9	87	96
Total	12	88	100

Frequentist question: $P(\text{Positive Test} \mid \text{No Disease})$

- Among the 96 healthy people, 9 tested positive
 - False positive rate: $\frac{9}{96} \approx 0.10$
If a person does *not* have the disease,
there is about a 10% chance of testing positive.
- Frequentist analysis focuses on: $P(\text{Data} \mid \text{Null Hypothesis})$
- A p-value is *not* the probability that the hypothesis is true.
(common misunderstanding)

Bayesian question: $P(\text{Disease} \mid \text{Positive Test})$

- Only 3 of the 12 positive tests are true positives.
 - Probability of actually having the disease: $\frac{3}{12} = 0.25$
Even with a positive test, disease probability is only 25%.
- Sensitivity : $P(+ \mid \text{Disease}) = \frac{3}{4} = 75\%$
- Specificity : $P(- \mid \text{No Disease}) = \frac{87}{96} \approx 90.6\%$

The Meaning of Probability

Probability describes uncertainty.

Frequentist:

- Probability is a **long-run frequency**.
- Describes what happens over many repeated trials.

Bayesian:

- Probability is a **degree of belief**.
- Measures how plausible something is given current information.

Bayesian thinking updates beliefs using evidence:

$$\text{Posterior} \propto \text{Prior} \times \text{Likelihood}$$

Probability is not only about randomness;
it is a mathematical language for learning under uncertainty.

Homework

Send in your answers before the next class.

Responses may be shared anonymously during class discussion.

Question 1

Think of a recent situation in which you changed your mind. Make a Bayesian knowledge-building diagram that includes your prior information, explain how new data helped you change your mind and your posterior conclusion.

Homework

Question 2

There are several data scientist openings at a heavily hyped company. Having read the job description, you know for a fact that you are qualified for the position (this is your data). Your goal is to ascertain whether you will actually be offered a position (this is your hypothesis).

- a. From the perspective of someone using frequentist thinking, what question is answered in testing the hypothesis that you'll be offered the position?
- b. Repeat part (a) within a Bayesian framework
- c. Which question would you rather have the answer to: the frequentist or the Bayesian? Explain.

Homework

Question 3

- a. Identify a topic that you know about (e.g., a sport, a school subject, music).
- b. Identify a hypothesis about this subject.
- c. How would your current expertise inform your conclusion about this hypothesis?
- d. Which framework are you employing here, Bayesian or frequentist?

Homework

Question 4

A university health center is considering whether to recommend a costly new screening test for a rare disease among students. Historical data suggest that only 2% of students have the disease. The test correctly identifies diseased students 95% of the time. The test falsely identifies healthy students as diseased 8% of the time. A student tests positive.

- a. What is the probability the student actually has the disease?
- b. Should the health center automatically recommend treatment after one positive test?
- c. What are the risks of a false positive? a false negative?
- d. How would the decision change if the disease were more common?

Homework

Question 5

Suppose two analysts disagree about the probability of a cyberattack before seeing evidence.

- Analyst A believes the probability is 1%.
- Analyst B believes the probability is 20%.

Both observe the same security alert.

- a. How can Bayesian reasoning explain why the analysts may still reach different conclusions?
- b. What role do priors play in decision-making under uncertainty?
- c. When is it reasonable for two people to maintain different posterior beliefs after observing the same data?
- d. How do costs and risks influence the final decision, beyond just the posterior probability?

Homework

Question 6

Your friend just became interested in Bayesian statistics. Explain the following to them:

- a. Why is Bayesian statistics useful?
- b. What are the similarities in Bayesian and frequentist statistics?